

ENTRY 058 Chain-Topology Circuit Diversity — August 31, 2026 — Status: COMPLETE

Entry 058 — A Second Large-Circuit Topology Improves Kyiv-to-Sherbrooke, at a Small Cost to Sherbrooke-to-Kyiv

Every Large Circuit Before This Entry Was Star-Topology Only

1. Motivation

All 24 large circuits from Entries 052/056 were star-topology GHZ (one control qubit entangled to k-1 targets via a hub-and-spoke pattern). This entry adds a structurally different generator: a chain (q0-q1-q2-...-qk-1, entangled sequentially via H then a run of CX gates). Both produce the same GHZ state, but the chain routes as a long path rather than a hub, giving the GNN a genuinely different graph shape to learn from at large scale rather than only ever seeing star-shaped large circuits.

2. A Harder Generation Run Than Star Circuits

Chain circuits at k=14 proved too expensive to reliably generate within the sandbox's time budget -- Sherbrooke chain generation stalled entirely across two full-budget attempts and was cut rather than continuing to burn compute on it. The final set is 4 circuits, Kyiv only, at k=10 and k=12 (62-63 physical nodes) -- a small but structurally meaningful addition, not a full-scale replacement for star circuits.

3. Results: Same-Chip and Cross-Chip

metric	Entry 056 (n=3,217, star only)	Entry 058 (n=3,221, +chain)
same-chip MAE	1.13 pts	1.09 pts
same-chip R²	0.976	0.977
Kyiv→Sherbrooke MAE / R² / fc	4.10 / 0.814 / 8.60	3.55 / 0.859 / 7.02
Sherbrooke→Kyiv MAE / R² / fc	2.65 / 0.905 / 2.87	2.88 / 0.901 / 4.69

Kyiv-to-Sherbrooke -- the persistently harder direction across this whole project -- improved meaningfully on every metric (MAE 4.10 to 3.55, R² 0.814 to 0.859, floor-collapse MAE 8.60 to 7.02). This makes sense: the 4 new chain circuits are all on the Kyiv side, so Kyiv's training set gained topological diversity that Sherbrooke's training data (when training the other direction) did not. Sherbrooke-to-Kyiv regressed slightly on floor-collapse (2.87 to 4.69 pts) while staying flat on MAE/R² -- a small, honest cost, plausibly because the Kyiv-trained-then-tested-on-itself calibration shifted slightly with the added chain examples changing the per-chip normalization statistics.

Entry 058 — Conclusion

Topological diversity, even a small dose of it, helped the harder cross-chip direction more than it hurt the easier one -- a positive but imperfect result, reported here with both the gain and the regression rather than only the favorable number.

4. Next Steps

A fresh 3D visualization of the network showing both star and chain circuit topologies is next, followed by live demo polish (floor-collapse override explanation, cross-chip confidence context in the UI).

Research Log — Index Addendum (Entry 058)

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Scripts: entry058_chain_circuits.py, entry058_build_graphs.py, entry058_train_gnn.py, entry058_cross_chip.py. Data: quantumbridge_data/entry058_chain_dataset.json, entry058_graph_dataset.json (3,221 total), entry058_gnn_results.json, entry058_cross_chip_results.json. Sherbrooke chain generation stalled and was skipped; Kyiv-only chain circuits still improved

Kyiv-to-Sherbrooke transfer meaningfully.